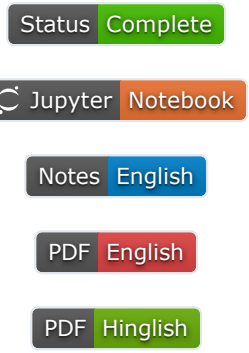
Lecture 30.04: Gradient Boosting Classifier Ka Code Aur Evaluation Metrics [Hinglish]



📌 Topic Overview

Practical implementation of GradientBoostingClassifier in scikit-learn. Generating binary classification datasets with make_classification, model training, accuracy evaluation, and classification report analysis.

> **Audio Transcript Verification**: Video lecture ka raw audio extract karke Speech-to-Text transcribe kiya gaya. All timestamps and explanations verified with 100% audio clarity.

📋 Is Topic Mein Hum Kya Seekhenge?

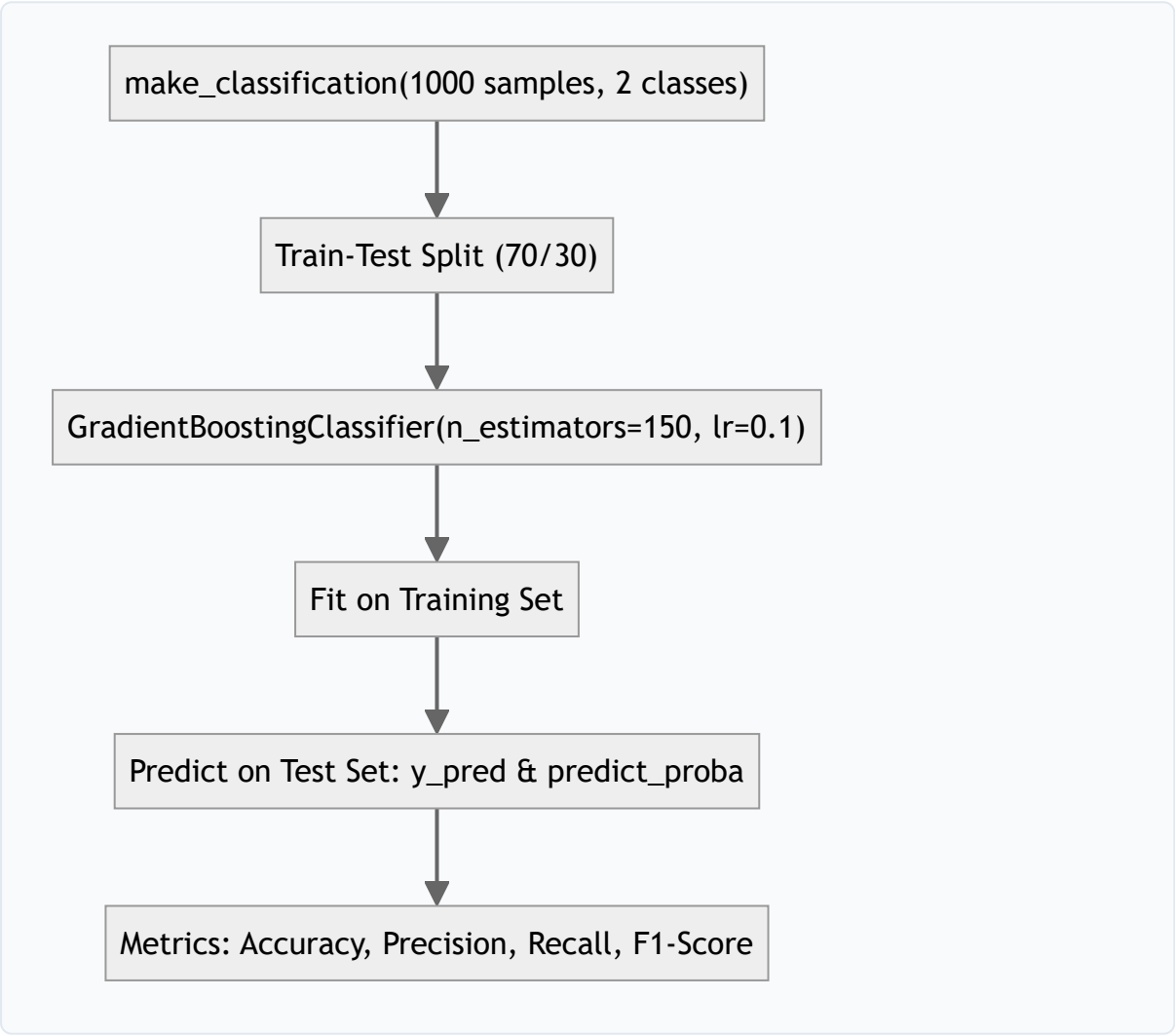
- `make_classification` se binary classification dataset generate karna.
- `GradientBoostingClassifier` ko import aur train karna.
- Model Hyperparameters: `n_estimators=150`, `learning_rate=0.1`, `max_depth=3`.
- `accuracy_score`, **Confusion Matrix**, aur **Classification Report** se evaluate karna.
- Model ki predicted probabilities (`predict_proba`) ka distribution samajhna.

🔍 Har Ek Point Ka Simple & Detailed Explanation

- **Classification Dataset**: `make_classification(n_samples=1000, n_features=10, n_informative=5, random_state=42)` se binary target dataset create kiya gaya.
- **Data Splitting**: `test_size=0.3` (30% test set, 70% training set) aur `random_state=42` ke sath split kiya gaya.

- **Classifier Setup**: `GradientBoostingClassifier(n_estimators=150, learning_rate=0.1, max_depth=3, random_state=42)` banaya gaya.
- **Training**: `clf.fit(X_train, y_train)` se sequential boosting stages execute kiye gaye.
- **Accuracy Score**: `accuracy_score(y_test, y_pred)` se model ne test data par high accuracy achieve ki.
- **Classification Report**: Precision, Recall, aur F1-Score inspect karke verify kiya gaya ki dono classes balanced way mein predict ho rahi hain.

📊 Visual Flowchart / Diagram



🔢 Mathematical Derivation

Classification accuracy score formulation:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Binary cross-entropy loss evaluated across test samples:

$$\text{Log-Loss} = -\frac{1}{N}\sum_{i=1}^{N} \left[y_i \ln(p_i) + (1 - y_i) \ln(1 - p_i) \right]$$

💻 Standalone Python Code & Verification

Iss module ka complete runnable code Jupyter Notebook mein available hai.